HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 121: Structured Programming Language

Batch: 29th | Semester: Spring 2026

Date: 06/09/2026 | **Final Examination** | Time: 2 Hours | Total Marks: 40

Answer any five questions. $5 \times 8 = 40$

1. (a) Analyze each of the program segments that follow and determine their outputs. CLO5 2×2=4

i.
```
for (int i = 1; i <= 5; i++) {
    for (int j = 1; j <= i; j++) {
        printf("* ");
    }
    printf("\n");
}
```

ii.
```
int x = 0;
for (int i = 1; i <= 4; i++) {
    for (int j = 1; j <= i; j++) {
        x++;
    }
}
printf("%d", x);
```

(b) Write a C program to print the following number pattern based on user input for the number of rows. The example pattern is when row = 5. CLO4 4

```
1
22
333
4444
55555
```

2. (a) Define array. Explain the need for array variables. CLO3 3

(b) A student has completed 6 subjects in a semester. The marks obtained in each subject are stored in an array. The university wants to determine the student's total and average marks to evaluate their overall academic performance. CLO4 5

Write a C program that takes the marks of 6 subjects as input, stores them in an array, and calculates and displays the total marks obtained and the average marks of the student.

3. (a) Briefly explain the working of the `strcmp()` function. Then, write a C program to input two strings, compare them using `strcmp()`, and display whether the strings are identical or different. CLO4 3

(b) A text-processing system needs to compress a sequence of repeated characters by replacing each group of consecutive identical characters with the character followed by the number of times it occurs consecutively. For example, the string `aaabbc` should be represented as `a3b2c1`. CLO4 5

Write a C program that takes a string containing lowercase letters as input

and generates its encoded form by displaying each character followed by the number of consecutive occurrences.

4. (a) Write the general format of a user-defined function definition in C and mention its main elements. CLO4 3

(b) A number-analysis program needs to determine whether a given positive integer is divisible only by 1 and itself. Write a C program using a user-defined function to perform this check and display the result. CLO4 5

5. (a) Distinguish between call by value and call by reference. CLO3 2

(b) Define recursive function. Write a C program using a recursive user-defined function to calculate and display the factorial of the input number. CLO4 6

6. (a) Briefly describe the reference(&) and dereference(*) operators used with pointer variables in C with examples. CLO4 2

(b) Find out the output of the following program segments – CLO5 3×2=6

i.
```
int a = 8;
int *p = &a;
*p = *p + 5;
a = a * 2;
*p -= 3;
printf("%d %d", a, *p);
```

ii.
```
int x = 7;
int *p = &x;
int **q = &p;
**q = **q + 4;
*p = *p - 2;
**q = **q * 3;
printf("x = %d\n", x);
```

iii.
```
int x = 20, y = 15; // Assume address of x and
y is 3000 and 4000
int *p, *q;
p = &x;
q = &y;
*p += 5;
*q = *p - *q;
p = q;
*p *= 2;
x -= 3;
printf("x = %d, y = %d, p = %d, q = %d\n", x,
y, p, q);
```

7. (a) State the differences between structure and union in C programming language. CLO4 2

(b) A university needs to maintain basic information about the courses offered in a semester. For each course, the system needs to store the course ID (int), section (char), and course fee (float). CLO4 6

Write a C program that defines a structure named **Course** to store these details. The program should take the information for 5 courses as input from the user using a loop and then display the details of all the courses.

HAMDARD UNIVERSITY BANGLADESH

Department of Electrical and Electronics Engineering
Bachelor of Science in Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT 125 **Course Title: Linear Algebra and Complex Variables**
Final Examination Semester: Spring 2026
Date: 09/13/2026 Total Time: 2:00 Hours **Total Marks: 40**

Answer any four (04) of the following questions: **4× 10 = 40**

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.		Questions	CLOs	Marks
Q1.	(a)	Define eigenvalues and eigenvectors with examples.	CLO1	[2]
	(b)	Find the eigenvalues and eigenvectors of the matrix	CLO3	[8]

$$A=\begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}.$$

Also find the matrix P that diagonalizes A and determine $P^{-1}AP$.

Q2.	(a)	Define Gaussian plane and properties of complex numbers.	CLO1	[4]
	(b)	If Z_1 and Z_2 are two complex numbers, then prove that $\lvert Z_1 - Z_2\rvert \le \lvert Z_1\rvert + \lvert Z_2\rvert$.	CLO3	[6]
Q3.	(a)	Define neighbourhoods, continuity, interior/exterior points.	CLO1	[4]
	(b)	Find an equation for a circle of radius 4 with centre at (-2, 1).	CLO2	[2]
	(c)	Find all the seventh roots of unity.	CLO3	[4]
Q4.	(a)	Define singularity with an appropriate example.	CLO1	[2]
	(b)	Prove that the function $f(z)=u(x,y)+iv(x,y)$, where	CLO3	[8]

$$f(z)=\begin{cases} \dfrac{x^3(1+i)-y^3(1-i)}{x^2+y^2}, & z\neq 0 \\ 0 & z=0 \end{cases}$$ is continuous and the Cauchy-Riemann

equations are satisfied at the origin. Yet $f'(z)$ does not exist there.

Q5.		State and prove the Cauchy-Riemann theorem for necessary and sufficient conditions for $f(z)$ to be analytic.	CLO2	[10]
Q6.	(a)	State the Milne-Thomson method.	CLO1	[3]
	(b)	Find a function v such that $f(z) = u + iv$ is analytic and express it in terms of z. If $u = 3x^2y+2x^2-y^3-2y^2$.	CLO3	[7]

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science and Engineering

Department of Computer Science and Engineering

BSC (Honors) in CSE

ENG 127: English Skills

Final Examination

Spring 2026

Batch: 29th

September 8, 2026

2nd Semester

Time: 2 Hours

Total Marks: 40

CLOs	Question/ Statement	Level	Marks
CLO1	**1. What is an essay and how many types of essays are there? Define with examples.**	Remember	5×1=5
CLO5	**2. Paraphrase the following text**	Apply	5×1=5

Computers have transformed nearly every aspect of modern life, from communication to education. Early machines were massive, slow, and limited in what they could do. Over the decades, engineers made computers smaller, faster, and far more powerful. Today, a device that fits in a pocket can perform tasks that once required entire rooms of equipment. This rapid progress continues to shape how people work, learn, and connect with one another.

CLO4	**3. Write a letter to the Registrar of your university requesting him to provide funding for an upcoming event.**	Create	08×1=08
CLO4	**4. Write a paragraph on any of the following topics:** a. If you were a president. b. Artificial Intelligence.	Create	10×1=10
CLO4	**5. Some people think that, as AI replaces human workers, it may create a chaotic world in future whereas others believe that, AI is making a better and easier world for us.** **To what extent do you agree or disagree?**	Create	12×1=12

HAMDARD UNIVERSITY BANGLADESH

Department of Electrical and Electronics Engineering
Faculty of Science, Engineering and Technology

Course Code: PHY 121 | **Course Title: Physics II**
Final Examination | Semester: Spring 2026
Date: 9/15/2026 | Total Time: 2:00 Hours | **Total Marks: 40**

Answer any four (04) out of the following questions: $4 \times 10 = 40$

[The figures in the right margin indicate full marks. Symbols are used have their usual meaning.]

			CLOs	Marks
Q1.	**(a)**	What is the main difference between the Faraday's law and Lenz's law?	CLO1	[2]
	(b)	Explain the direction of the induced current according to Faraday's law with a neat sketch.	CLO2	[3]
	(c)	Deduce the mathematical expression for Hall voltage, $V_H = \frac{BI}{pw}$	CLO2	[5]
Q2.	**(a)**	What are the difference between inertial and non-inertial frame of reference?	CLO1	[2]
	(b)	Derive the Galilean Transformation Equations and write it drawbacks.	CLO2	[3]
	(c)	Explain the Michelson-Morley Experiment in briefly.	CLO2	[5]
Q3.	**(a)**	What are the Postulates of Special Relativity?	CLO1	[3]
	(b)	Derive the expression for time dilation using the postulates of special relativity.	CLO2	[4]
	(c)	The Hypersonic Technology Vehicle 2 (HTV-2) is an experimental rocket vehicle capable of traveling at 21,000 km/h (5830 m/s). If an electronic clock in the HTV-2 measures a time interval of exactly 1-s duration, what would observers on Earth measure the time interval to be?	CLO3	[3]
Q4.	**(a)**	Derive the mass-energy equivalence equation $\boldsymbol{E = mc^2}$ and explain its physical significance.	CLO2	[4]
	(b)	Derive the expressions for relativistic momentum and total energy of a particle moving with velocity (v). Hence, show that $E^2 = P^2C^2 + m_0{}^2c^4$.	CLO2	[6]
Q5.	**(a)**	What is the De-Broglie Hypothesis?	CLO1	[2]
	(b)	Derive the expression of De-Broglie wavelength.	CLO2	[5]
	(c)	Calculate the velocity and de Broglie wavelength of an α-particle of energy 1 keV. Given mass of a-particle 6.68×10^{-27} kg.	CLO3	[3]

Q6 **(a)** Define the following terms: CLO1 [3]

i. Wave packet
ii. Phase velocity
iii. Group velocity.

(b) Show that the de Broglie group velocity associate with the wave packet is equal to the velocity of the particle. CLO2 [7]

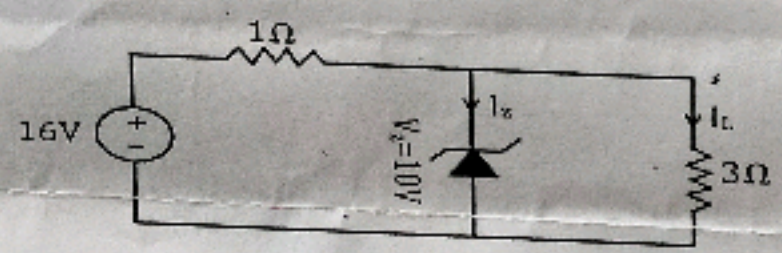
HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
EEE127: Electronic Devices and Circuits

Batch: 29th | Semester: Spring 2026
Date: 1-9-2026 | **Final Examination** | Time: 2 Hours
Total Marks: 40

Answer any five questions of the followings. $5 \times 8 = 40$

1.	a)	What is breakdown in diode? Classify breakdown in details.	CLO1	4
	b)	Describe the working principle of zener diode. How can a zener diode withstand breakdown?	CLO1	4
2.	a)	Can a regular P-N junction diode withstand breakdown? Justify your answer with proper explanation.	CLO1	3
	b)	Determine the value of Iz from the following circuit :	CLO2	5



3.	a)	How can a zener diode be used as a voltage regulator?	CLO1	3
	b)	For the given circuit calculate the range of R_L and I_L that will result in V_{RL} being maintained at 10V.	CLO2	5

